
Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix, spanning hydropower, wind, and solar technologies.

Source: [Our World in Data](#) — Ember (2026); Energy Institute (2025)



75% of Global GHG Emissions **Still Come from Fossil Fuels**

Since the Industrial Revolution, fossil fuels have dominated energy systems. The consequences are measurable and urgent.

75%

of global GHG emissions from fossil fuel energy

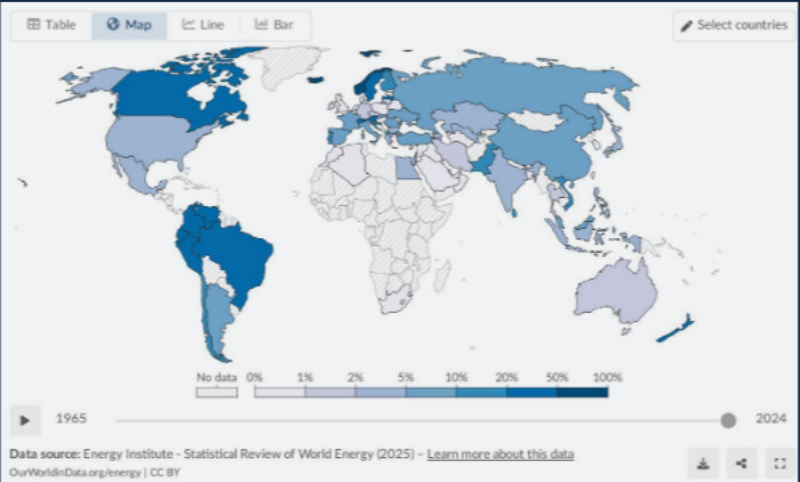
5M+

premature deaths per year from air pollution

1/7

of primary energy now from renewables

Renewables Now Supply 14% of Global Primary Energy



Share of primary energy from renewables by country, 2024 (substitution method)

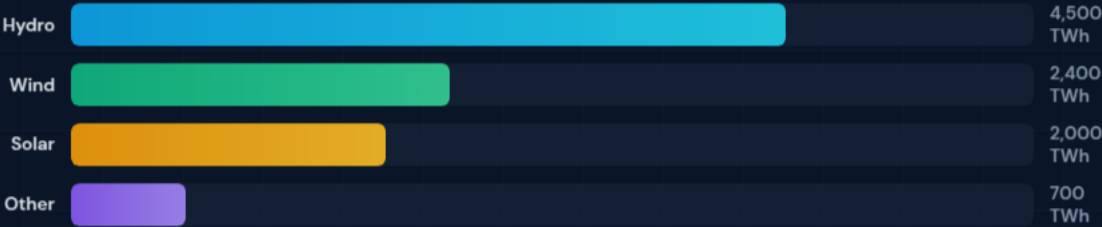
Primary energy encompasses electricity, transport, and heating combined. Because transport and heating are harder to decarbonize, the renewable share in total energy is lower than in electricity alone.

-14% of global primary energy from renewables (2024, substitution method)

Norway, Brazil, and New Zealand lead; Middle East and parts of Asia remain below 5%.

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

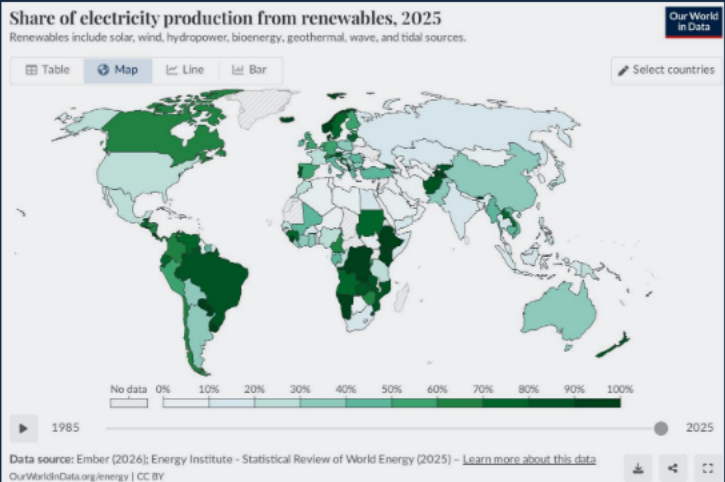
Wind and solar combined now nearly match hydropower's output — and they are growing much faster.



TECHNOLOGY	GENERATION (TWH)	SHARE
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other	~700	~7%
Total	~9,600	100%

Approximate global renewable electricity generation, 2024. Wind + solar = ~46% of renewable electricity.

Almost **One-Third** of Global Electricity Now Comes from Renewables



Share of electricity from renewables by country, 2025

REGION / COUNTRY	RENEWABLE SHARE
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%
World average	~30%

Several African and Middle Eastern countries have renewable shares below 10%.

Hydropower Remains the Largest Renewable Source at ~50% of Generation

Hydropower has been generating electricity at scale for over a century. It remains the single largest renewable electricity source, accounting for roughly half of all renewable generation.

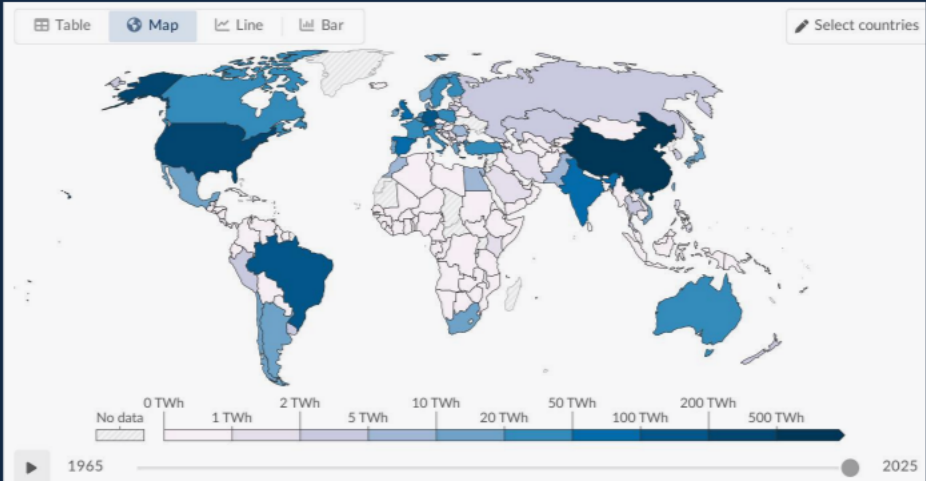
~4,500

TWh generated (2024)

~47%

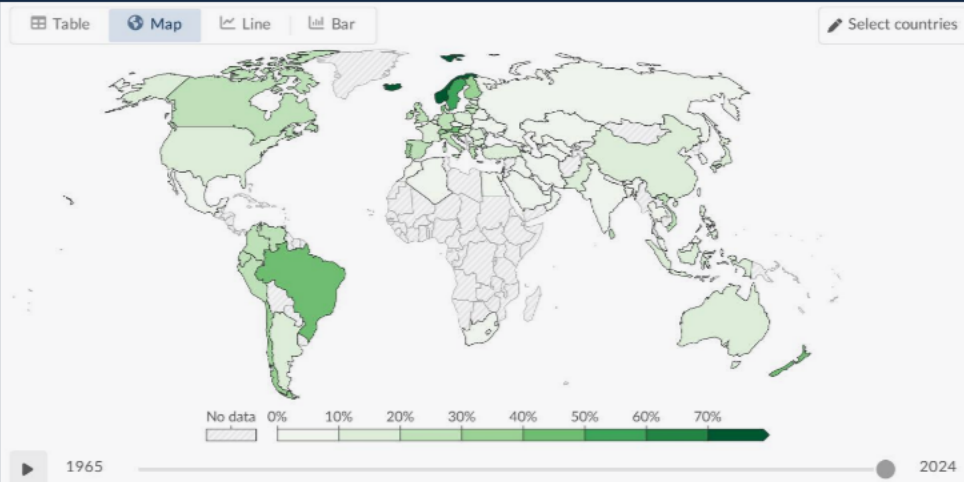
of renewable electricity

Top producers: China, Brazil, Canada. Growth potential is limited compared to wind and solar.



Hydropower generation by country, 2025

Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind power generation by country, 2025

-2,400 TWh

Wind generation
~25% of renewable electricity

-2,000 TWh

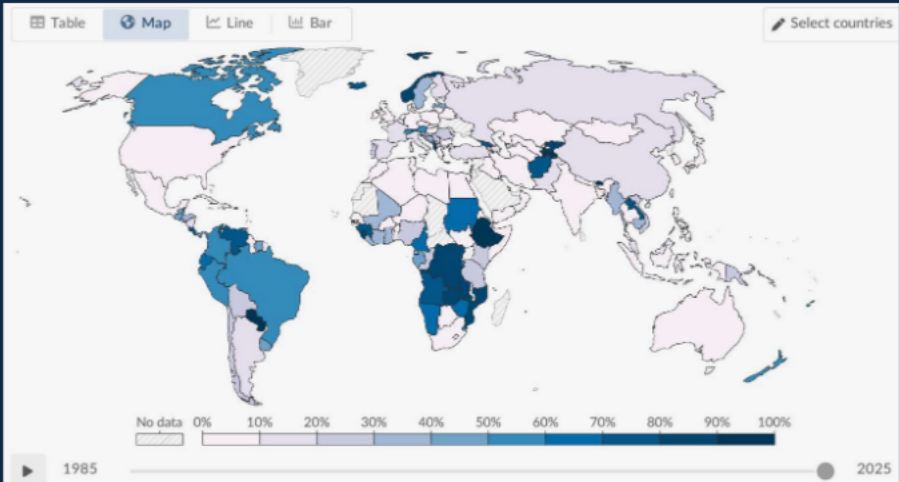
Solar generation
~21% of renewable electricity

Solar is the fastest-growing source, with dramatic growth since 2010 driven by falling costs and policy support. Wind and solar combined now nearly match hydropower output.

-46%

Wind + Solar share of renewable electricity — closing in on hydro's 47%

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia & Latin America Lead



Share of electricity from hydropower by country, 2025

65-98%
Scandinavia, Brazil, Canada

<10%
Sub-Saharan Africa, Middle East

The transition is highly uneven. Scandinavian countries and Brazil benefit from abundant hydropower. Denmark and Germany have invested heavily in wind and solar. Meanwhile, much of Sub-Saharan Africa has minimal renewable infrastructure, and many Middle Eastern nations remain almost entirely fossil-fuel dependent.

Key Takeaways: Renewables Are Growing Fast but **the Transition Must Accelerate**

- **~14% of global primary energy** and **~30% of global electricity** now come from renewables — a significant milestone but far from sufficient for climate targets.
- **Hydropower remains dominant**, generating roughly half of all renewable electricity, but its growth potential is limited compared to wind and solar.
- **Wind and solar are the fastest-growing energy technologies** in history; combined, they now produce nearly as much electricity as hydropower.
- **Stark regional disparities persist**: Scandinavia and Latin America exceed 60–90% renewable electricity, while much of the Middle East and Sub-Saharan Africa remain below 10%.
- **Transport and heating lag far behind electricity** in the renewable transition, keeping the overall primary energy share much lower than the electricity share.

The transition is underway, but the pace must accelerate dramatically to meet climate targets. Every sector — electricity, transport, and heating — needs rapid decarbonization.